

Development, simulation & implementation of precoding techniques for satellite links

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Master's dissertation submitted in order to obtain the degree of Master of Science in Computer Science Engineering

Academic year 2025-2026

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Ghent, May 2026

Wannes Baes

Acknowledgements

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by

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Department of Telecommunications and Information Processing

Abstract — Laboris magna irure quis sit et veniam voluptate duis elit esse do excepteur. Officia laboris adipisicing laboris duis ut cillum aliquip. Do veniam aute non do cillum sit pariatur consectetur adipisicing qui. Lorem Lorem veniam minim exercitation cupidatat tempor do aute dolore. Nostrud adipisicing tempor quis qui fugiat sint quis ut aute eiusmod Lorem enim veniam. Elit labore Lorem do proident amet anim exercitation culpa enim cillum qui. Aliquip reprehenderit commodo labore minim laborum mollit ut veniam in irure. Minim sit in non aliqua ut elit veniam irure pariatur non velit consequat. Cupidatat veniam occaecat culpa eu irure aute consectetur dolore culpa deserunt quis officia do. Aliqua aliqua eu velit excepteur culpa ea dolore occaecat. Ea est cillum nulla ea cupidatat esse adipisicing ut. Cillum amet esse ex ex enim elit excepteur ullamco in qui aliqua Lorem veniam. Sunt consectetur aliquip commodo exercitation pariatur id tempor adipisicing aliqua. Fugiat cillum ea irure laborum consectetur ipsum. Excepteur laboris eu veniam ex ad consequat esse id ullamco ea cupidatat ad nisi. Aute nisi velit excepteur adipisicing dolore pariatur sunt. Dolore mollit cupidatat esse sint ipsum est nulla nisi eu cupidatat excepteur sint cillum. Culpa irure magna culpa eiusmod ut nisi ipsum veniam elit est occaecat veniam. Sunt adipisicing eu minim qui ea laboris ullamco ut. Aliquip occaecat in occaecat ut tempor dolore sunt consequat qui cillum reprehenderit. Voluptate commodo cupidatat ullamco sint aliquip veniam consequat do. Nisi velit excepteur aute laboris aliqua fugiat minim occaecat quis non incididunt.

Keywords — Laboris magna, Laboris magna, Laboris magna, Laboris magna

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- Do not confuse “imply” and “infer”.
- The prefix “non” is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the “et” in the Latin abbreviation “et al.”.

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An excellent style manual for science writers is [1].

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TABLE I
TABLE TYPE STYLES

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Fig. 1. Example of a figure caption.

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ACKNOWLEDGMENT

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REFERENCES

- [1] A. I. Perez-Neira, M. A. Vazquez, M. B. Shankar, S. Maleki, and S. Chatzinotas, “Signal processing for high-throughput satellites: Challenges in new interference-limited scenarios,” *IEEE Signal Processing Magazine*, vol. 36, no. 4, pp. 112–131, 2019.

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Symbols and Notations

Abbreviations

BS	base station
CS	circularly symmetric
IBR	information bit rate
i.i.d.	independent and identically distributed
MI	mutual information
MIMO	multiple-input multiple-output
MU	multi-user
pdf	probability density function
RX	receiver
SISO	single-input single-output
SNR	signal-to-noise ratio
SR	sum rate
SU	single-user
SVD	singular value decomposition
TX	transmitter
UT	user terminal
WSR	weighted sum rate

Mathematical Notations

TO DO

System Parameters

N_T Number of transmit antennas at the BS

N_R Number of receive antennas at each UT

N_S Number of data streams per UT

K Number of UTs

Symbols

TO DO

Chapter 1

Introduction

Part I

Background Material

Chapter 2

Communication Systems

This chapter establishes the system models and information-theoretic framework that underlie the analysis presented throughout this work. We begin by describing the general architecture of a digital communication system and then specialize to the system model adopted in this work. Finally, we review the fundamental information-theoretic concepts required to analyze the performance of these systems.

2.1 Digital Communication Systems

A general digital communication system transfers digital information from a source to a destination over a physical transmission medium, referred to as the channel. The corresponding block diagram is shown in Fig. 2.1.

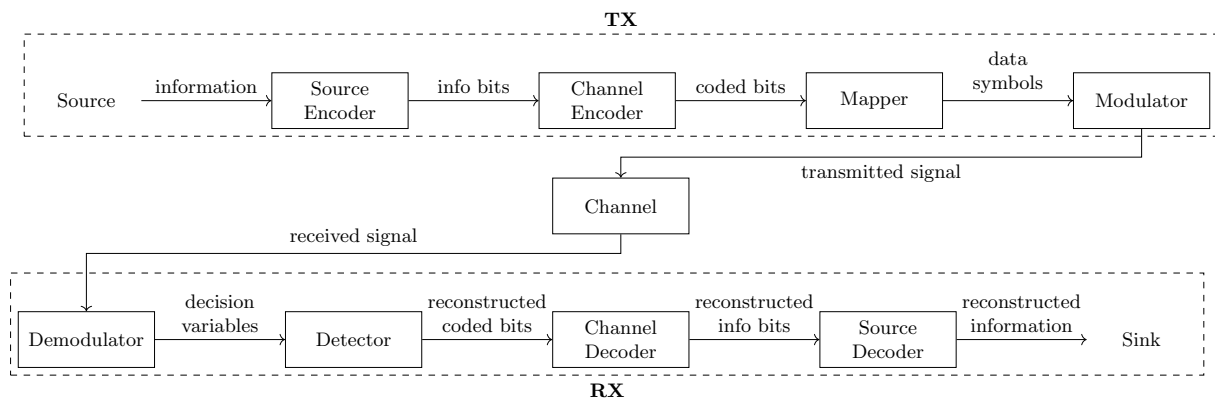


Figure 2.1: Block diagram of a general digital communication system.

At the transmitter (TX), the information sequence is first processed by a source encoder to remove redundancy, followed by a channel encoder that introduces controlled redundancy to enable error detection. The resulting coded bit sequence is partitioned into blocks of m_c bits, each of which is mapped onto a complex-valued symbol drawn from a constellation $\mathcal{X}$ of size 2^{m_c} . The modulator then converts the symbol sequence into a baseband signal suitable for transmission.

During transmission, the channel distorts the signal due to attenuation, multipath propagation, and additive noise, such that the received signal differs from the transmitted signal.

At the receiver (RX), the demodulator converts the received signal into a sequence of decision variables. These are processed by the detector to obtain estimates of the originally transmitted symbols, which are then mapped back to coded bits. Finally, the channel and source decoders reconstruct the original information sequence.

In this work, we consider a simplified version of this general architecture in order to focus on the aspects most relevant to our study. Source coding and decoding are omitted, and the information is assumed to be available as a sequence of independent and identically distributed (i.i.d.) bits without prior compression or digitization. Channel coding and decoding are also not considered, such that the analysis is restricted to uncoded transmission.

The transmission medium is assumed to be a wireless channel affected by fading and additive white Gaussian noise. Furthermore, the channel is considered to be frequency-flat, meaning that its response remains constant over the entire signal bandwidth. The specific channel models used in this work are introduced in the relevant chapters.

2.2 MIMO Communication Systems

When both the TX and the RX are equipped with multiple antennas, the system is referred to as a multiple-input multiple-output (MIMO) system.

Compared to a single-antenna link, the multiple antennas provide additional spatial degrees of freedom that can be exploited to increase data throughput, improve reliability, or achieve a trade-off between both. Spatial multiplexing techniques enable significant capacity gains by transmitting multiple independent data streams simultaneously over different antennas. A well-known example is the BLAST architecture [1]. Alternatively, spatial diversity can be achieved through space-time coding schemes, which distribute symbols across both space and time to improve robustness against fading. A classical example is the Alamouti scheme [2]. Throughout this work, the focus is on spatial multiplexing and, in particular, on how the simultaneous transmission of multiple data streams over a shared channel can be made most efficient through precoding.

2.2.1 SU-MIMO System Model

In the point-to-point case, where a single TX with N_T antennas communicates with a single RX with N_R antennas, the system is called a single-user (SU) MIMO system. A block diagram is shown in Fig. 2.2.

The discrete-time complex baseband input-output relation of a SU-MIMO Gaussian channel is characterized by

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad (2.1)$$

where $\mathbf{x} \in \mathbb{C}^{N_T}$ is the transmitted signal vector, $\mathbf{y} \in \mathbb{C}^{N_R}$ is the received signal vector, $\mathbf{H} \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix, and $\mathbf{n} \in \mathbb{C}^{N_R}$ is the additive noise vector. The (n_r, n_t) -th entry of $\mathbf{H}$ represents the complex channel coefficient between transmit antenna n_t and receive

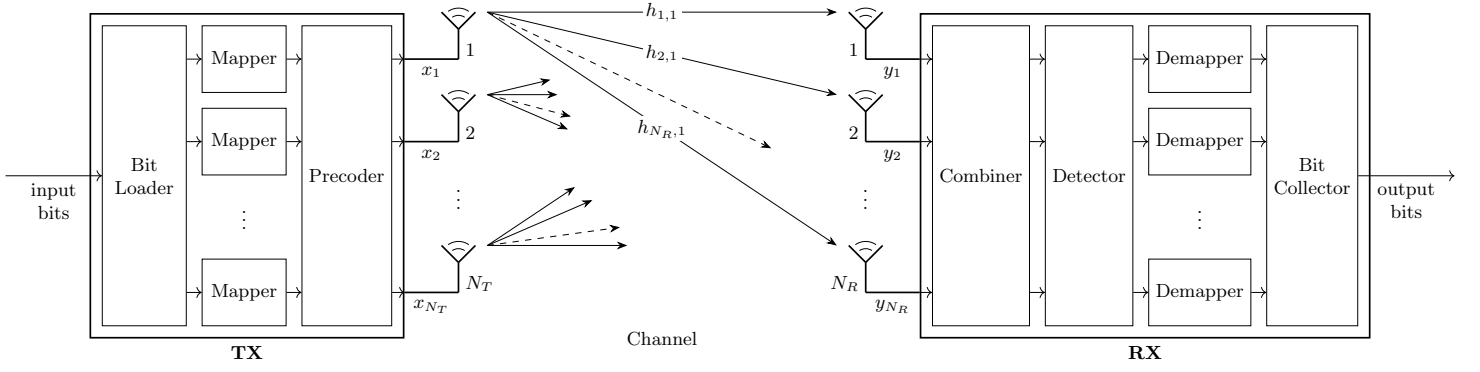


Figure 2.2: Block diagram of a SU-MIMO system.

antenna n_r . The noise vector is circularly symmetric (CS) complex Gaussian distributed, i.e. $\mathbf{n} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, N_0 \mathbf{I}_{N_R})$, such that its entries are spatially independent, each with an expected power of N_0 . The time index is omitted for brevity.

At the TX, the information bits are divided into N_S streams and mapped onto data symbols, which are collected in the vector $\mathbf{a} \in \mathbb{C}^{N_S \times 1}$. Then, the transmitted signal is obtained by mapping the N_S data symbol streams onto the N_T transmit antennas according to a linear precoding scheme, represented by a matrix $\mathbf{F} \in \mathbb{C}^{N_T \times N_S}$. The transmitted signal can therefore be expressed as

$$\mathbf{x} = \mathbf{F} \mathbf{a}. \quad (2.2)$$

The data symbols are assumed to be zero-mean, mutually uncorrelated, and normalized, such that $\mathbb{E}[\mathbf{a}\mathbf{a}^H] = \mathbf{I}_{N_S}$. Throughout this work, the power allocation is assumed to be incorporated into the precoding matrix, unless explicitly stated otherwise. Under this convention, the total transmit power is constrained by

$$\mathbb{E}[\|\mathbf{x}\|^2] = \text{tr}(\mathbf{F}^H \mathbf{F}) \leq P_t. \quad (2.3)$$

At the RX, the N_R received signals are linearly combined to obtain the decision-variable vector $\mathbf{z} \in \mathbb{C}^{N_S \times 1}$. The combiner is represented by a matrix $\mathbf{G} \in \mathbb{C}^{N_S \times N_R}$. The decision variables can therefore be expressed as

$$\mathbf{z} = \mathbf{G} \mathbf{y}. \quad (2.4)$$

Based on $\mathbf{z}$, the detector forms estimates of the transmitted data symbols, which are then mapped back to bits.

Substituting (2.2) into (2.1) and applying (2.4) yields the relation between the transmitted data symbols and the decision variables:

$$\mathbf{z} = (\mathbf{G} \mathbf{H} \mathbf{F}) \mathbf{a} + \mathbf{G} \mathbf{n}, \quad (2.5a)$$

$$= \mathbf{T} \mathbf{a} + \mathbf{n}_{\mathbf{G}}, \quad (2.5b)$$

where $\mathbf{T} \in \mathbb{C}^{N_S \times N_S}$ is the MIMO transfer matrix and $\mathbf{n}_{\mathbf{G}}$ represents spatially correlated Gaussian noise. It is evident that, whenever the transfer matrix is not diagonal, the data symbols of different

streams interfere with each other. The design of the precoding and combining matrices is therefore essential to suppress this inter-stream interference and improve overall system performance.

2.2.2 MU-MIMO System Model

When multiple receivers are served simultaneously by a single multi-antenna transmitter, the system is referred to as a multi-user (MU) MIMO system. In this context, the transmitter is called the base station (BS) and the individual receivers are referred to as user terminals (UTs). A block diagram is shown in Fig. 2.3.

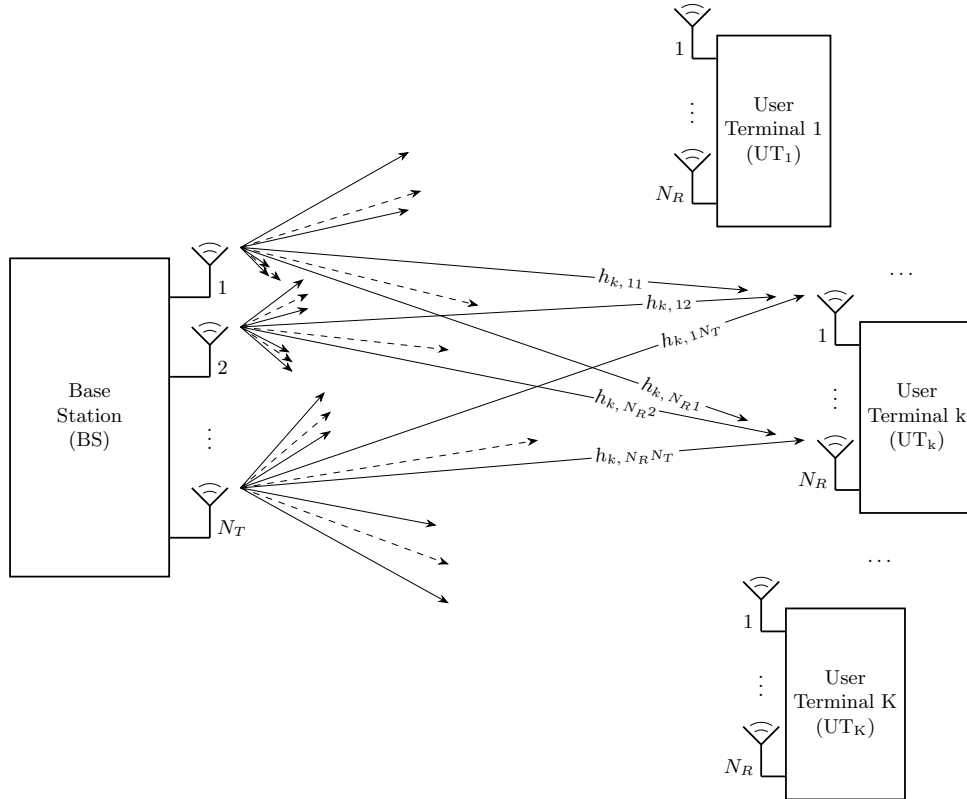


Figure 2.3: Block diagram of a MU-MIMO system.

Throughout this work, the BS is assumed to be equipped with more antennas than all UTs combined, and each UT is assumed to receive at most as many data streams as it has receive antennas. For notational convenience, all UTs are assumed to receive the same number of data streams, denoted by N_S . In practical scenarios where this does not hold, N_S is defined as the maximum number of data streams intended for any single UT. The data symbol vectors intended for a UT receiving fewer streams are padded with zeros.

Analogous to the SU-MIMO case, the communication between the BS and a certain UT_k at a particular time instant is characterized by

$$\mathbf{y}_k = \mathbf{H}_k \mathbf{x} + \mathbf{n}_k, \quad k = 1, \dots, K, \quad (2.6)$$

where $\mathbf{H}_k \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix that describes the link between the BS and UT_k .

The transmitted signal vector $\mathbf{x} \in \mathbb{C}^{N_T}$ is now a function of the data symbol vectors intended for different UTs. The BS assigns a dedicated precoding scheme $\mathbf{F}_k \in \mathbb{C}^{N_T \times N_S}$ to the data symbol vector $\mathbf{a}_k$ intended for UT_k, and transmits the superposition of all precoded streams:

$$\mathbf{x} = \sum_{k=1}^K \mathbf{F}_k \mathbf{a}_k. \quad (2.7)$$

Since the data symbols intended for different UTs are assumed to be uncorrelated, the total transmit power constraint can now be formulated as

$$\mathbb{E}[\|\mathbf{x}\|^2] = \sum_{k=1}^K \text{tr}(\mathbf{F}_k^H \mathbf{F}_k) \leq P_t. \quad (2.8)$$

Applying the combining scheme $\mathbf{G}_k \in \mathbb{C}^{N_S \times N_R}$ at UT_k to the received signal vector and substituting (2.7) into (2.6) yields

$$\mathbf{z}_k = \underbrace{(\mathbf{G}_k \mathbf{H}_k \mathbf{F}_k)}_{\text{desired signal}} \mathbf{a}_k + \underbrace{\sum_{k'=1, k' \neq k}^K (\mathbf{G}_k \mathbf{H}_k \mathbf{F}_{k'}) \mathbf{a}_{k'}}_{\text{inter-user interference}} + \underbrace{\mathbf{G}_k \mathbf{n}_k}_{\text{noise}}. \quad (2.9)$$

The key difference from the SU-MIMO scenario is the presence of inter-user interference. Each UT receives not only its desired signal but also the signals intended for all other UTs. Suppressing this interference through precoder design at the BS is an important challenge in MU-MIMO systems.

The overall input-output relation of the MU-MIMO Gaussian channel can be written compactly as

$$\mathbf{z} = (\mathbf{G} \mathbf{H} \mathbf{F}) \mathbf{a} + \mathbf{G} \mathbf{n}, \quad \text{with} \quad (2.10)$$

$$\mathbf{z} = \begin{bmatrix} \mathbf{z}_1 \\ \vdots \\ \mathbf{z}_K \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \mathbf{G}_1 & & \\ & \ddots & \\ & & \mathbf{G}_K \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_1 \\ \vdots \\ \mathbf{H}_K \end{bmatrix}, \quad \mathbf{F} = [\mathbf{F}_1 \quad \dots \quad \mathbf{F}_K], \quad \mathbf{a} = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_K \end{bmatrix}.$$

where $\mathbf{G} \in \mathbb{C}^{KN_S \times KN_R}$ denotes the compound combining matrix, $\mathbf{H} \in \mathbb{C}^{KN_R \times N_T}$ the compound channel matrix, and $\mathbf{F} \in \mathbb{C}^{N_T \times KN_S}$ the compound precoding matrix.

2.3 Information Theory Concepts

In this section, we introduce the information-theoretic concepts needed for the analysis of MIMO communication systems. We first discuss mutual information, then define the achievable rate and capacity for SU-MIMO channels, and finally extend these concepts to the MU scenario.

2.3.1 Mutual Information

Consider a single use of a SU-MIMO Gaussian channel, as described by (2.1). The transmitter generates a random vector $\mathbf{x} \in \mathbb{C}^{N_T}$ according to the input distribution with probability density function (pdf) $p(\mathbf{x})$, while the receiver observes the random vector $\mathbf{y} \in \mathbb{C}^{N_R}$. The relation between the channel input $\mathbf{x}$ and the channel output $\mathbf{y}$ is probabilistic and is fully characterized by the channel law, described by the conditional pdf $p(\mathbf{y} | \mathbf{x})$.

The number of information bits transmitted per channel use is referred to as the information bit rate (IBR). The mutual information (MI) between $\mathbf{x}$ and $\mathbf{y}$ quantifies how much information the received vector $\mathbf{y}$ provides about the transmitted vector $\mathbf{x}$. In other words, it represents the average reduction in the uncertainty about $\mathbf{x}$ obtained from observing $\mathbf{y}$. It is given by

$$I(\mathbf{x}; \mathbf{y}) = \mathbb{E} \left[\log_2 \left(\frac{p(\mathbf{y} | \mathbf{x})}{p(\mathbf{y})} \right) \right], \quad (2.11)$$

where the expectation is taken with respect to the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ and $p(\mathbf{y})$ is the marginal distribution of $\mathbf{y}$.

According to Shannon's channel coding theorem [3], reliable communication is possible if the transmission IBR is strictly smaller than the MI between $\mathbf{x}$ and $\mathbf{y}$. Here, reliable communication means that the decoding error probability can be made arbitrarily small by using sufficiently long channel codes.

2.3.2 Achievable Rate and Capacity

For a given input distribution $p(\mathbf{x})$ and channel law $p(\mathbf{y} | \mathbf{x})$, any IBR r smaller than the MI is said to be achievable. The MI is thus a supremum (least upper bound) on the achievable rate.

The capacity C of a SU-MIMO channel is the largest achievable rate. Equivalently, it is the highest spectral efficiency for which reliable communication is possible. It is defined as the maximum MI between $\mathbf{x}$ and $\mathbf{y}$ over all possible input distributions $p(\mathbf{x})$:

$$C = \max_{p(\mathbf{x})} I(\mathbf{x}; \mathbf{y}). \quad (2.12)$$

Typically, the maximization is subject to the transmit power constraint, as formulated in (2.3).

In [4], the capacity for a SU-MIMO Gaussian communication system is derived. First, it is shown that under the total transmit power constraint, the MI is maximized in case of Gaussian signaling, i.e. the channel input $\mathbf{x}$ is CS complex Gaussian distributed. Then, the mutual information between $\mathbf{x}$ and $\mathbf{y}$ for a given covariance matrix $\mathbf{C}_x$ is obtained as

$$I(\mathbf{x}; \mathbf{y}) = \log_2 \left(\det \left(\mathbf{I}_{N_R} + \mathbf{C}_n^{-1} \mathbf{H} \mathbf{C}_x \mathbf{H}^H \right) \right). \quad (2.13)$$

The capacity then results from maximizing (2.13) over all positive semidefinite covariance matrices $\mathbf{C}_x$ that satisfy the transmit power constraint:

$$\begin{aligned} C &= \max_{\mathbf{C}_x} \log_2 (\det (\mathbf{I}_{N_R} + \mathbf{C}_n^{-1} \mathbf{H} \mathbf{C}_x \mathbf{H}^H)) \\ \text{s.t. } &\mathbf{C}_x \succeq 0, \quad \text{tr}(\mathbf{C}_x) \leq P_t. \end{aligned} \quad (2.14)$$

2.3.3 Rate Region and Sum Capacity

In contrast to the SU scenario, where the IBR is a scalar r , a MU-MIMO channel is characterized by a rate tuple $(r_1, \dots, r_K)$, where each element r_k denotes the IBR of a single UT. The rate region $\mathcal{R}$ is the set of all rate tuples whose individual IBRs are simultaneously achievable under the given system constraints. The rate region is a convex set, since any convex combination of achievable rate tuples is itself achievable through time-sharing between the corresponding transmission strategies. An example rate region for $K = 2$ users is shown in Fig. 2.4.

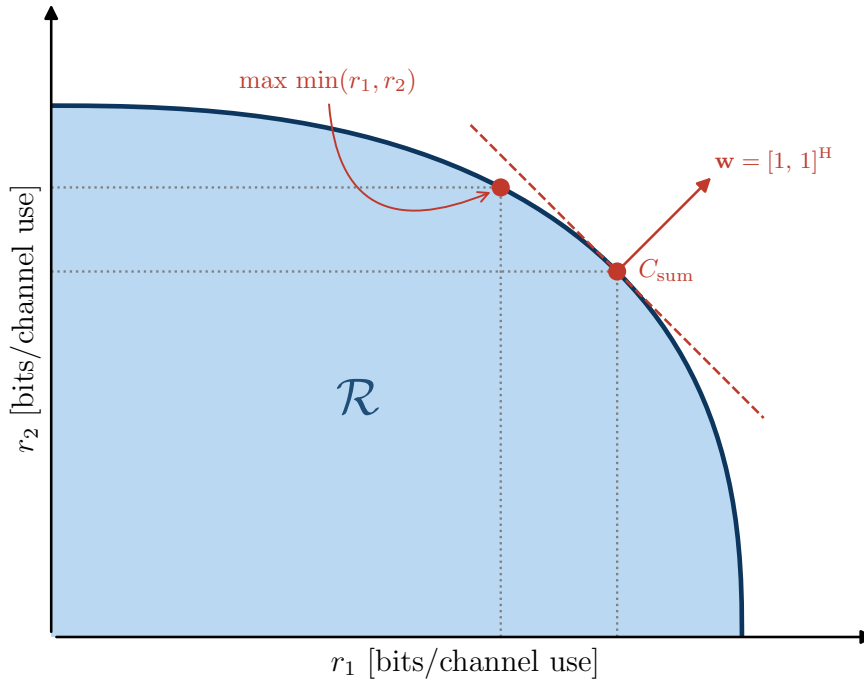


Figure 2.4: Example of a rate region $\mathcal{R}$ for a MU-MIMO system with $K = 2$ users.

The sum capacity point is the boundary point whose outward normal is the all-ones vector. The max-min point is given by the intersection of the boundary with the line $r_1 = r_2$. It corresponds to a fair operating point where both users achieve the same IBR.

The rate region captures the trade-offs inherent to multi-user transmission: increasing the IBR of one user may reduce the rates achievable by others due to interference and shared resources. It is therefore desirable to operate on the boundary of the rate region, where no user's rate can be increased without reducing that of another.

The sum capacity C_{sum} is the maximum achievable sum rate (SR), i.e., the sum of all individual IBRs:

$$C_{\text{sum}} = \max_{(r_1, \dots, r_K) \in \mathcal{R}} \sum_{k=1}^K r_k. \quad (2.15)$$

More generally, we define the weighted sum rate (WSR) as

$$\text{WSR} = \sum_{k=1}^K w_k r_k, \quad (2.16)$$

where $w_k \geq 0$ is the weight assigned to UT_k . The full boundary of the rate region can then be traced by maximizing the WSR over all achievable rate tuples for varying weight vectors $\mathbf{w} = (w_1, \dots, w_K)$. The sum capacity corresponds to the special case where all weights are equal to one. The maximization in both cases is typically subject to the total transmit power constraint (2.8).

In [5, 6, 7], the sum capacity of the MU-MIMO Gaussian channel is derived by exploiting the uplink–downlink duality: the downlink broadcast channel and the uplink multiple-access channel yield the same rate region under the same total transmit power constraint. It therefore suffices to solve the uplink problem, which is more tractable.

In the uplink channel, all K UTs transmit simultaneously, with UT_k sending a vector $\mathbf{x}_k^{\text{up}} \in \mathbb{C}^{N_R}$ to the BS. The MI between the transmitted vectors $\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}}$ and the received vector $\mathbf{y}$, given input covariance matrices $\mathbf{C}_{\mathbf{x},k}$, is defined as

$$I(\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}}; \mathbf{y}) = \mathbb{E} \left[\log_2 \left(\frac{p(\mathbf{y} | \mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}})}{p(\mathbf{y})} \right) \right], \quad (2.17)$$

where the expectation is taken with respect to the joint pdf of $(\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}})$ and $\mathbf{y}$, and $p(\mathbf{y})$ is the marginal pdf of $\mathbf{y}$. It is further obtained as

$$I(\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}}; \mathbf{y}) = \log_2 \left(\det \left(\mathbf{I}_{N_T} + \mathbf{C}_n^{-1} \sum_{k=1}^K \mathbf{H}_k \mathbf{C}_{\mathbf{x},k}^{\text{up}} \mathbf{H}_k^H \right) \right). \quad (2.18)$$

The uplink SR is achievable if and only if the sum of the IBRs of all UTs is strictly smaller than the MI as given in (2.18). Therefore, the sum capacity of the downlink channel is obtained by maximizing (2.18) over all positive semidefinite covariance matrices satisfying the total transmit power constraint and applying the duality result:

$$\begin{aligned} C_{\text{sum}} &= \max_{\mathbf{C}_{\mathbf{x},1}^{\text{up}}, \dots, \mathbf{C}_{\mathbf{x},K}^{\text{up}}} \log_2 \left(\det \left(\mathbf{I}_{N_T} + \mathbf{C}_n^{-1} \sum_{k=1}^K \mathbf{H}_k \mathbf{C}_{\mathbf{x},k}^{\text{up}} \mathbf{H}_k^H \right) \right) \\ \text{s.t.} \quad &\sum_{k=1}^K \text{tr}(\mathbf{C}_{\mathbf{x},k}^{\text{up}}) \leq P_t \quad \text{and} \quad \mathbf{C}_{\mathbf{x},k}^{\text{up}} \succeq 0 \quad \forall k = 1, \dots, K. \end{aligned} \quad (2.19)$$

Chapter 3

Precoding for SU-MIMO systems

3.1 Capacity Achieving Precoding

Consider the SU-MIMO system model introduced in Section 2.2.1 together with the associated capacity optimization problem in (2.14). We first derive a more explicit expression for the channel capacity and then show that this capacity can be achieved by means of linear precoding. Parts of this section follow a structure similar to that in Chapter 3 of [8].

3.1.1 SU-MIMO Capacity

The singular value decomposition (SVD) (see Appendix A) of the channel matrix yields

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H, \quad (3.1)$$

where $\mathbf{U} \in \mathbb{C}^{N_R \times N_R}$ and $\mathbf{V} \in \mathbb{C}^{N_T \times N_T}$ are unitary matrices, and $\mathbf{\Sigma} \in \mathbb{R}^{N_R \times N_T}$ is a rectangular diagonal matrix which contains the singular values of $\mathbf{H}$ on its main diagonal in decreasing order. The columns of $\mathbf{U}$ and $\mathbf{V}$ contain the left and right singular vectors of $\mathbf{H}$, respectively.

Now define the transformed variables $\tilde{\mathbf{y}} = \mathbf{U}^H \mathbf{y}$, $\tilde{\mathbf{x}} = \mathbf{V}^H \mathbf{x}$, and $\tilde{\mathbf{n}} = \mathbf{U}^H \mathbf{n}$. Since multiplication by a unitary matrix preserves the distribution of CS complex Gaussian noise, $\tilde{\mathbf{n}}$ has the same distribution as $\mathbf{n}$. Moreover, because $\mathbf{U}$ and $\mathbf{V}$ are unitary, the transformations between $(\mathbf{x}, \mathbf{y})$ and $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ are bijective. The original channel from (2.1) can therefore be represented equivalently as

$$\tilde{\mathbf{y}} = \mathbf{\Sigma} \tilde{\mathbf{x}} + \tilde{\mathbf{n}}, \quad (3.2)$$

which is obtained by applying the SVD of the channel matrix and performing a change of variables (see Appendix C). The channel in (3.2) is equivalent to the original channel in the sense that the transformation preserves the mutual information between input and output. Consequently, both channels have the same capacity.

The channel matrix $\mathbf{H}$ has exactly as many non-zero singular values $\{\sigma_i\}$ as its rank, denoted as R_H . Written component-wise, the equivalent channel becomes

$$\tilde{y}_i = \sigma_i \tilde{x}_i + \tilde{n}_i \quad \text{for } i = 1, \dots, R_H, \quad (3.3)$$

with $\tilde{y}_i = \tilde{n}_i$ for $R_H < i \leq N_R$. The equivalent channel decomposes into R_H independent parallel scalar Gaussian subchannels, each with its own channel gain σ_i . These subchannels and their gains are referred to as eigen subchannels and eigenmodes, respectively. Since the components $\tilde{x}_i$ for $R_H < i \leq N_T$ have no effect on the output, they carry no information and can be set to zero.

For each eigen subchannel, the mutual information (MI) is maximised by a zero-mean, CS complex Gaussian input [4]. Under this choice, the optimisation problem (2.14) simplifies to

$$\begin{aligned} C &= \max_{\{\mathbb{E}[|\tilde{x}_i|^2]\}} \sum_{i=1}^{R_H} \log_2 \left(1 + \mathbb{E}[|\tilde{x}_i|^2] \frac{\sigma_i^2}{N_0} \right) \\ \text{s.t. } &\sum_{i=1}^{R_H} \mathbb{E}[|\tilde{x}_i|^2] \leq P_t. \end{aligned} \quad (3.4)$$

This maximization problem is solved using the Lagrange multiplier method (see Appendix B). The variances of the channel input need to be chosen according to the waterfilling solution [9]:

$$\mathbb{E}[|\tilde{x}_i|^2] = \left(\mu_w - \frac{N_0}{\sigma_i^2} \right)^+, \quad (3.5)$$

where the water level μ_w is found iteratively to satisfy the transmit power constraint with equality.

The resulting capacity is given by

$$C = \sum_{i=1}^{R_H} \log_2 \left(\max(1, \mu_w \frac{\sigma_i^2}{N_0}) \right). \quad (3.6)$$

3.1.2 Optimal Precoder Design

In this section, we design a linear precoder and combiner that together achieve the capacity of the SU-MIMO channel. We assume that the data symbol vector is CS complex Gaussian distributed and spatially uncorrelated, and restrict ourselves to the case where the number of data streams is not fixed in advance, but can be chosen freely as part of the design.

For an arbitrary precoding matrix $\mathbf{F}$ and combining matrix $\mathbf{G}$, the data processing inequality [10] yields $I(\mathbf{x}; \mathbf{y}) \geq I(\mathbf{a}; \mathbf{z})$, where the MI between the transmitted data symbols $\mathbf{a}$ and the decision variables $\mathbf{z}$ is given by:

$$I(\mathbf{a}; \mathbf{z}) = \log_2 \left(\det(\mathbf{I}_{N_s} + (\mathbf{G} \mathbf{C}_n \mathbf{G}^H)^{-1} (\mathbf{G} \mathbf{H} \mathbf{F}) \mathbf{C}_a (\mathbf{G} \mathbf{H} \mathbf{F})^H) \right) \quad (3.7a)$$

$$= \log_2 \left(\det \left(\mathbf{I}_{N_s} + \frac{1}{N_0} (\mathbf{G} \mathbf{G}^H)^{-1} (\mathbf{G} \mathbf{H} \mathbf{F}) (\mathbf{G} \mathbf{H} \mathbf{F})^H \right) \right). \quad (3.7b)$$

Expression (3.7a) follows directly from (2.13) and the structural similarity between (2.1) and (2.5). Expression (3.7b) then follows from (3.7a) upon substituting $\mathbf{C}_a = \mathbf{I}_{N_s}$ and $\mathbf{C}_n = N_0 \mathbf{I}_{N_R}$.

Motivated by the capacity derivation in the previous subsection, we choose the columns of $\mathbf{F}$ and the rows of $\mathbf{G}$ as the R_H most dominant right and left singular vectors of $\mathbf{H}$, respectively. These are precisely the singular vectors associated with the non-zero singular values of $\mathbf{H}$. The number of data streams is therefore chosen equal to the rank of the channel matrix. In addition, the precoder incorporates the power allocation through a diagonal matrix $\mathbf{P}$. More specifically, we set

$$\mathbf{F} = \mathbf{V}_{N_s} \sqrt{\mathbf{P}} \quad \text{and} \quad \mathbf{G} = \mathbf{U}_{N_s}^H. \quad (3.8)$$

It follows from (3.8) that $I(\mathbf{a}; \mathbf{z}) = I(\mathbf{x}; \mathbf{y})$ for an appropriate choice of the power allocation matrix $\mathbf{P}$. This result can be verified in a brute-force way by substituting the particular precoder and combiner matrices from (3.8) into (3.7) and replacing the channel matrix by its SVD (see Appendix C).

An alternative and more elegant approach consists in substituting the particular precoder and combiner into (2.5) and comparing the resulting expression with the equivalent channel given in (3.2). This yields

$$\mathbf{z} = \mathbf{U}_{N_s}^H \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H \mathbf{V}_{N_s} \sqrt{\mathbf{P}} \mathbf{a} + \mathbf{U}_{N_s}^H \mathbf{n} \quad (3.9a)$$

$$= \mathbf{\Sigma}_{N_s} \sqrt{\mathbf{P}} \mathbf{a} + \tilde{\mathbf{n}}, \quad (3.9b)$$

where $\mathbf{\Sigma}_{N_s} \in \mathbb{R}^{N_s \times N_s}$ contains the R_H non-zero singular values of $\mathbf{H}$ on its diagonal. Comparing (3.9) with (3.2), one can readily verify that $I(\mathbf{a}; \mathbf{z}) = I(\mathbf{x}; \mathbf{y})$, provided that the power allocated to each stream is set according to the waterfilling solution. Hence, transmission with the precoder and combiner from (3.8) achieves the capacity of the SU-MIMO channel.

Denoting the s th element of the decision variable vector $\mathbf{z}$ by z_s , one can write (3.9) component-wise as

$$z_s = \sigma_s \sqrt{p_s} a_s + \tilde{n}_s \quad \text{for } s = 1, \dots, N_s. \quad (3.10)$$

This expression shows that the precoder and combiner in (3.8) diagonalize the SU-MIMO transfer matrix from (2.5). The original SU-MIMO channel is decoupled into $N_s = R_H$ independent parallel scalar single-input single-output (SISO) Gaussian subchannels with gains σ_s , which coincides exactly with the decomposition of the equivalent channel in (3.3). A visualization of this decomposition is shown in Fig. 3.1.

As a result, the individual data streams become statistically independent. Detection therefore reduces to symbol-by-symbol processing rather than joint detection of the full decision-variable vector. This leads to a significant reduction in receiver complexity, especially for systems with a large number of antennas and data streams.

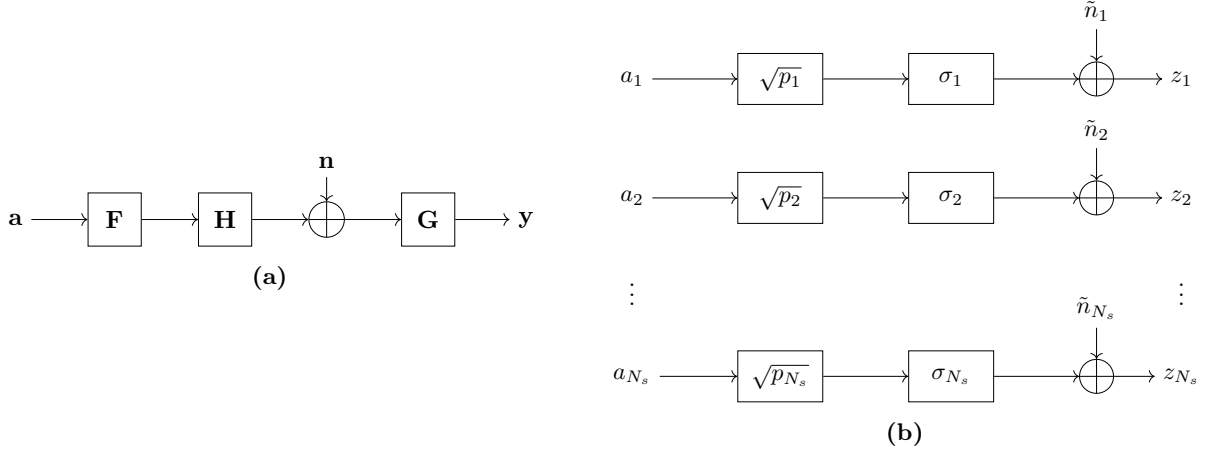


Figure 3.1: A block diagram of the original SU-MIMO channel (a) and its decomposition into parallel SISO Gaussian eigen subchannels (b) by means of the optimal precoder and combiner.

No general closed-form expression exists for the optimal water level μ_w , and hence for the resulting power allocation. Nevertheless, the optimal power allocation is fully characterized by

$$\sum_{i=1}^{N_s} p_i = P_t \quad \text{and} \quad p_i = \left(\mu_w - \frac{N_0}{\sigma_i^2} \right)^+ . \quad (3.11)$$

Since the total allocated power is a monotonically increasing function of the water level, the latter can be determined numerically by means of a bisection method. A more efficient approach exploits the fact that the number of active streams (those receiving strictly positive power) is a discrete quantity. First, the number of active streams is determined via a binary search in at most $\lceil \log_2 N_s \rceil$ iterations, after which the water level follows in closed form. Finally, the power allocated to each stream is computed. The complete procedure is summarized in Algorithm 1, in Appendix B. Note that this approach relies on the fact that the set of candidate streams is finite and therefore does not extend in the same form to waterfilling algorithms for power allocation across the frequency spectrum of a frequency-selective channel.

The resulting solution admits a natural waterfilling interpretation, which also motivates the name of the algorithm. For each eigen subchannel, the quantity $\gamma_i^{-1} = N_0/\sigma_i^2$ can be interpreted as a floor level, while the water level μ_w is shared across all subchannels. Power is poured into each eigen subchannel container, so to speak, until the common water level is reached. Consequently, no power is assigned to eigen subchannels whose floor level exceeds the water level. The structure of the optimal power allocation is illustrated in Fig. 3.2 for a SU-MIMO system with four transmit and four receive antennas operating at an signal-to-noise ratio (SNR) of 5 dB: eigen subchannels with stronger channel conditions, i.e. larger gains σ_i , receive more power than weaker ones.

At extremely low SNR, the waterfilling solution corresponds to the eigenbeamforming power allocation strategy, which concentrates all available power on the strongest eigen subchannel. As a result, only a single data stream is transmitted in this SNR regime. This behavior is illustrated in Fig. 3.3a. Conversely, at an extremely high SNR, the optimal power allocation strategy reduces to the uniform power allocation, which distributes power equally across all eigen subchannels,

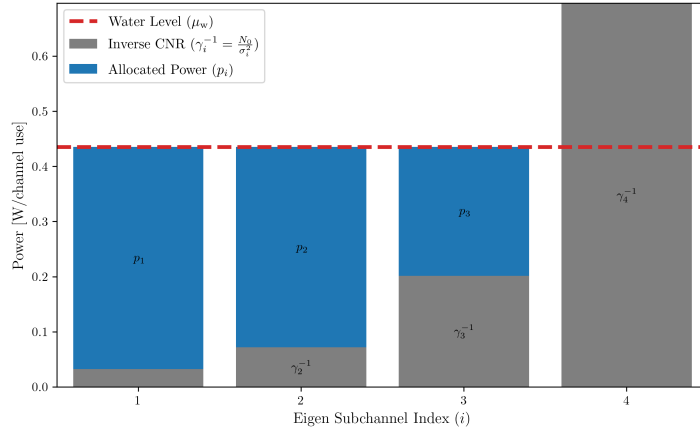


Figure 3.2: Optimal power allocation for a SU-MIMO system with 4 transmit and 4 receive antennas and a total transmit power P_t of 1 W per channel use. The system operates at an SNR of 5 dB.

as illustrated in Fig. 3.3b. It is worth noting that uniform power allocation does not require knowledge of the current channel matrix, in contrast to the general waterfilling solution.

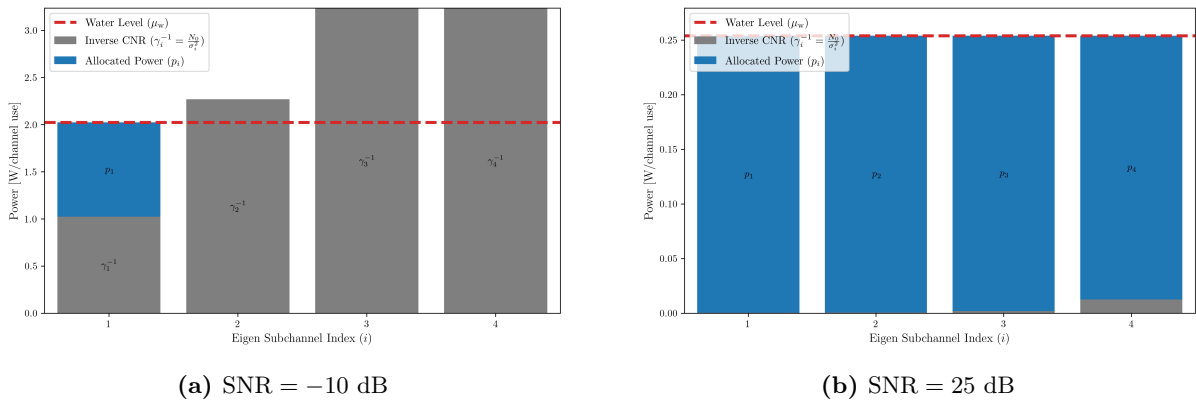


Figure 3.3: Waterfilling power allocation for a SU-MIMO system with 4 transmit and 4 receive antennas at different SNR extremes. The maximum transmit power P_t is 1 W per channel use.

3.2 Joint Design for Linear Precoder and Combiner

3.2.1 Generalized WMMSE Design

3.2.2 Maximum Information Rate Design

3.2.3 Quality of Service Design

Part II

Precoding for Satellite Communications

Part III

Appendices

Appendix A

Singular Value Decomposition

Appendix B

Lagrangian Optimization

Algorithm 1 Waterfilling Algorithm for power allocation across eigen subchannels

Require: The total transmit power P_t ; the SNR ratios $\gamma_i = \sigma_i/N_0$ for $i = 1, \dots, N_s$, sorted in decreasing order.

Ensure: The optimal power allocation p_i for $i = 1, \dots, N_s$.

Step 1: Determine the optimal number of active streams

- 1: $N_s^{\text{LB}}, N_s^{\text{UB}} \leftarrow 0, N_s$
- 2: **while** $N_s^{\text{UB}} - N_s^{\text{LB}} > 1$ **do**
 - Update the number of active streams:
- 3: $N_s^{\text{iter}} \leftarrow \lceil (N_s^{\text{UB}} + N_s^{\text{LB}})/2 \rceil$
 - Compute the minimum required power to activate N_s^{iter} streams:
- 4: $P_{\min} \leftarrow \sum_{i=1}^{N_s^{\text{iter}}} (1/\gamma_{N_s^{\text{iter}}} - 1/\gamma_i)$
 - Update the bound on the number of active streams:
- 5: **if** $P_{\min} \leq P_t$ **then** $N_s^{\text{LB}} \leftarrow N_s^{\text{iter}}$ **else** $N_s^{\text{UB}} \leftarrow N_s^{\text{iter}}$
- 6: **end while**

Step 2: Compute the power allocated to each stream

- Fill the N_s^{LB} active stream containers:
- 7: $p_i \leftarrow 1/\gamma_{N_s^{\text{LB}}} - 1/\gamma_i$ for $i = 1, \dots, N_s^{\text{LB}}$
 - Distribute the remaining power equally among the active streams:
 - 8: $p_i \leftarrow p_i + (P_t - \sum_{i=1}^{N_s^{\text{LB}}} p_i) / N_s^{\text{LB}}$ for $i = 1, \dots, N_s^{\text{LB}}$
 - Set the power of the inactive streams to zero:
 - 9: $p_i \leftarrow 0$ for $i = N_s^{\text{LB}} + 1, \dots, N_s$

Return p_i for $i = 1, \dots, N_s$

Appendix C

Extended Calculations

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